

ECOTRACK AI

Cloud-Native Sustainability Platform



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1. Executive Summary & Mission Statement

EcoTrack AI is a state-of-the-art sustainability platform designed to empower individuals and organizations to combat climate change. By integrating robust Domain-Driven Design (DDD) with Clean Architecture, local heuristic intelligence forecasting, and visual mechanics, EcoTrack AI converts raw emissions tracking into actionable, gamified, and highly accessible paths towards a carbon-neutral lifestyle.

Our core mission is to bridge the gap between individual carbon awareness and sustained behavioral modification. We achieve this by providing accurate calculators, interactive dashboards, projection analysis, and community-centered challenges that turn sustainability tracking into a rewarding journey.

2. Project Scope & Functional Deliverables

The EcoTrack AI platform consists of six key functional areas:

- **Carbon Calculator**

Enables users to log daily data points in five key categories: Transportation (km), Energy (kWh), Food (meals), Waste (kg), and Water (liters) using standard conversion factors.

- **Dashboard Analytics**

Displays sustainability scores (0-100), summarizes monthly categories via clean charts, and tracks rolling weekly emission trends.

- **AI Recommendation Engine**

Examines user history to detect carbon drivers and outputs prioritized, high-impact suggestions tailored to their lifestyle.

- **Goal Tracker with AI Predictions**

Allows users to set carbon reduction targets. Utilizes logistic model forecasting to predict the probability of successfully hitting targets.

- **Gamified Challenges**

Encourages collective action. Users can join and complete community-driven campaigns to earn points and practice sustainable habits.

- **Admin Management Panel**

Provides administrators the ability to update category emission multipliers dynamically and audit system-wide security logs.

3. System Architecture & Tech Stack

The platform is built on Clean Architecture principles to enforce separation of concerns, testability, and maintainability. The codebase is cleanly split into frontend client, backend API, and container infrastructure layers.

Layer	Core Technologies	Architectural Purpose
Frontend Client	React, TypeScript, Vite, Vanilla CSS	Client application, state management, SPA routing
Frontend Web Server	Nginx	Static asset serving, secure proxy to API routes (/api)
Backend Services	Node.js (Express), TypeScript	REST endpoints, domain logic services, middleware rules
Database Layer	Prisma ORM, PostgreSQL	Database mapping, migration logging, transactional consistency
Infrastructure hosting	Docker, GCP Cloud Run	Serverless auto-scaling microservices, containerization
Secrets Management	GCP Secret Manager	Encrypted storage of DB URLs and JWT signing secrets
CI/CD Pipeline	GitHub Actions	Automated linting, typescript compile checks, builds & deployment

4. Database Entity Relationships

The relational database schema mapped via Prisma consists of the following key models:

- User: Manages profiles, credentials, role-based controls (USER/ADMIN), and demographics.
- RefreshToken: Manages secure sliding auth sessions.
- CarbonLog: Keeps itemized daily entries of logged footprints (category, sub-category, units, kg CO2).
- EmissionFactor: Custom multiplier values for converting units into CO2 equivalents.
- Goal: Target limits with start/end boundaries and target thresholds.
- Challenge & UserChallenge: Join-state trackers for gamified campaigns.
- Notification & AuditLog: Activity feeds and security audit trails.

5. Core Mathematical & AI Formulas

Formula A: Carbon Equivalent Calculation

Every logged quantity is immediately converted into kilograms of CO2 equivalent (kg CO2e) using dynamically configured multipliers standard in IPCC guidelines:

$$\text{Emission} = \text{Value} * \text{Multiplier}$$

Formula B: Dynamic Sustainability Scoring (0 - 100)

The sustainability score is evaluated out of 100 points, encouraging balanced behaviors:

1. Carbon Profile (40 pts): Computed based on daily emissions compared to a baseline (12 kg/day):
$$\text{Score} = \text{Max}(0, 1 - \text{DailyAverage} / 24) * 40$$
2. Green Habits (20 pts): 10 points for active recycling, 10 points for renewable energy adoption.
3. Goal Commitment (25 pts): Scaled by goal success rate ($\text{CompletionRate} * 25$).
4. Challenges (15 pts): Scaled by active participations ($\text{ActiveChallenges} * 5$, capped at 15).

Formula C: Goal Success Probability Projection

To forecast if a user will meet their carbon limits, the platform calculates projected ending emissions based on their current rate, and routes the margin through a logistic activation curve:

$$P(\text{Success}) = 1 / (1 + e^{-(k * \text{normalized_margin})})$$

Where $k = 5$ (steepness multiplier), and $\text{normalized_margin} = (\text{Target} - \text{Projected}) / \text{Target}$.

This projects realistic probabilities and outputs custom warnings if success drops below 40%.

Formula D: Linear Regression Trend Forecasting

For datasets with 3+ days of entries, the AI engine runs a linear regression ($y = mx + c$) to calculate slope and intercepts. This projects future rolling averages at 30, 90, and 365 days with dynamic 95% confidence intervals based on data variance:

$$\text{Confidence Interval} = \text{ProjectedSum} \pm (\text{StdDev} * 1.96 * \text{Sqrt}(30))$$

6. CI/CD Pipeline & Deployment Strategy

Deployment is fully automated from source to cloud using a multi-stage GitHub Actions runner:

1. Test & Validation: Installs project files, runs type verification, and executes the Jest suite.
2. Authentication: Exchanges OIDC tokens via Google Workload Identity Federation (WIF).
3. Build & Register: Compiles Docker builds and pushes images to GCP Artifact Registry.
4. Secrets Retrieval: Queries Secret Manager dynamically to fetch production Supabase credentials.
5. Deploy: Deploys backend and frontend to auto-scaling Google Cloud Run instances. Passes backend URLs dynamically to Nginx configuration files to proxy API calls.

7. Generative Design Asset Prompt

To establish a premium branding identity, a design layout was generated using the following prompt:

"A premium, state-of-the-art UI dashboard mockup for a sustainability app called EcoTrack AI. The interface should have a dark mode design with glassmorphic cards, vibrant green and teal accent colors, and display charts, a sustainability score (e.g., '78'), carbon footprint categories, and modern icons. The background should be a subtle, smooth gradient of dark gray to deep navy. Extremely high detail, clean typography, look like a professional SaaS dashboard, no device frames, no mockup templates."

The generated image represents the landing visual for stakeholders and is embedded in the report's title page.

End of Report. Ready for Production Release.